

# Sci-kit Learn Recreation and Implementation

CSE299-sec-5

Group-5

Salman Shafi-2211386042

Jarinah Tasnim-2121026642

Jesmin Akter-2111382642

Sazid Rahman Bhuiyan-2131267042

## Week 6 ML Library Implementation - Project Update

Week 6 ML Library Implementation has been successfully completed with comprehensive advanced features and automation systems that significantly enhance the library's functionality and usability. We implemented **Principal Component Analysis (PCA)** for dimensionality reduction with configurable variance ratio selection, whitening capabilities, and automatic component determination, enabling efficient data compression and visualization for high-dimensional datasets like digits and complex feature spaces. **Advanced feature selection utilities** were developed including **SelectKBest** for univariate statistical feature ranking using F-score classification tests and **Recursive Feature Elimination (RFE)** for iterative feature removal based on model importance, both integrated seamlessly with existing preprocessing workflows. A **professional sklearn-compatible Pipeline system** was created allowing chaining of multiple preprocessing steps including scaling, dimensionality reduction, feature selection, and final estimators in a single coherent workflow with parameter passing and cross-validation support. **Comprehensive model persistence functionality** was implemented using pickle serialization with metadata storage, enabling users to save trained models with configuration details and load them for future predictions or analysis. **Four automation bash scripts** were developed for complete workflow automation including classification, regression, clustering, and comprehensive pipeline testing with color-coded output, performance timing, result organization, and automatic report generation. The **command-line interface was extensively enhanced** with Week 6 specific arguments including PCA parameters (components, whitening), feature selection options (method, number of features), pipeline activation, and model persistence paths. **Advanced metrics display** was implemented for dimensionality reduction tasks showing explained variance ratios, reconstruction errors, singular values, and component analysis alongside existing classification and regression metrics. **Comprehensive integration testing** was conducted across all datasets (Iris, Wine, Digits, Boston) with various combinations of preprocessing steps, ensuring robust performance and compatibility with existing Week 1-5 implementations. The complete system now supports **complex machine learning workflows** with professional-grade preprocessing pipelines, advanced feature engineering capabilities, model management systems, and automated testing frameworks, positioning the library as a **production-ready tool** for educational, research, and practical machine learning applications.

